END-TO-END AI AUTOMATION PROJECT

Teaching Materials Generator

Project Documentation and Handover Guide

An n8n workflow that turns one lesson topic into a differentiated teaching-package draft and emails it to the teacher as a PDF for review.

Project	Teaching Materials Generator (n8n workflow: "teaching materials generator")
Document version	1.0
Date	31 October 2026
Workflow export	teaching materials generator final.json (20 nodes, status: Active)
Platform	n8n, self-hosted (localhost:5678), published through an ngrok tunnel
AI model	Google Gemini (models/gemini-3.1-flash-lite), 6 AI Agent nodes
Prepared by	Dr Doaa Hafez

Related documents: Technical Workflow Specification (n8n), Prompt Library, and the sample output "Teaching material package" (PDF).

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1. Executive Summary

Business problem

Teachers spend a long time preparing separate materials for the same lesson: an explanation, a student handout, versions for different ability levels, a dyslexia-friendly version, practice exercises, a quiz with an answer key, and visual aids. Preparing all of this by hand for every lesson is repetitive, and the differentiated and accessible versions are the parts most often skipped for lack of time.

Solution

The Teaching Materials Generator is an automated workflow. A teacher fills in one web form (subject, grade, topic, learning objective, duration, language, and optional details). Six AI agents then generate a 12-section teaching package. The workflow checks that all required section headings are present, converts the package to PDF, and emails it to the teacher. No manual content-generation or formatting steps are required after submission; the teacher must still review the material before classroom use.

What the teacher receives

A single PDF containing sections A to L: learning objectives, teacher explanation, student handout, three differentiated levels, a dyslexia-friendly version, activities, practice exercises, discussion questions, a quiz with answer key, a summary, visual-content prompts, and accessibility and teacher guidance.

Key facts

Item	Detail
Trigger	n8n Form Trigger (web form, 11 fields)
AI	6 sequential AI Agent nodes using Google Gemini
Quality control	If node confirms all 12 section headings (A to L) are present before anything is sent
Output	PDF attached to an email sent to the address entered in the form
Verified run	Topic "Living organisms and non-living things": the PDF package (234 KB) was delivered by email, sent under the name "TeachWise assistant"
Human review	Required. The teacher must verify facts, quiz answers and the answer key before classroom use

2. Overview

2.1 Goal and business value

Goal: automatically produce a differentiated teaching-package draft from a single form submission and deliver it by email for teacher review.

- Saves lesson-preparation time: one submission replaces the manual writing of up to 12 separate resources.
- Makes differentiation and accessibility routine: Beginner, Intermediate, Advanced and dyslexia-friendly versions are produced every time.
- Consistent structure: every package uses the same 12 headings, so teachers know where to find each resource.
- Delivery in minutes, directly to the teacher's inbox, with no manual copying or formatting.

2.2 Users and scope

Role	What they do
Teacher	Opens the form link, enters lesson details, receives the PDF by email, and reviews it before use.
Workflow owner	Maintains the workflow in n8n, keeps credentials valid, monitors executions, and updates prompts.
Operations / support	Keeps n8n and ngrok running, handles escalations, follows the runbook in Section 10.

In scope: form intake, AI generation of 12 sections, section validation, PDF creation, email delivery, failure notification.

Out of scope: fact-checking of AI content (done by the teacher), storage of generated packages, curriculum-standard verification, student data.

2.3 Technology stack

Component	Technology	Role
Orchestration	n8n (self-hosted)	Runs the workflow and holds credentials
AI	Google Gemini via Google Gemini Chat Model nodes	Generates the teaching content
Agents	n8n AI Agent nodes (6)	One agent per group of sections, each with its own prompt
PDF conversion	html2pdf.app API (HTTP Request node)	Converts the HTML page to a PDF file
Email	Gmail (OAuth2)	Sends the PDF or the failure message
Public access	ngrok tunnel	Makes the local form reachable by teachers

3. Design and Planning

This section records the design decisions, following the Discovery, Decomposition and Architecture phases from the course.

3.1 Problem definition

Question	Answer
What is automated?	Creation of a full set of lesson resources for one topic.
Who needs it?	Primary-school teachers (the prompts are written for primary science).
What categories of output?	12 sections, A to L (table in 3.3).
Which details can the teacher control?	11 form fields, including language, available resources, learner needs and assessment requirements.
Baseline for comparison	Manual preparation time per lesson. To be measured by the project owner and recorded in Section 9.

3.2 Workflow map: trigger, steps, branches, outcomes

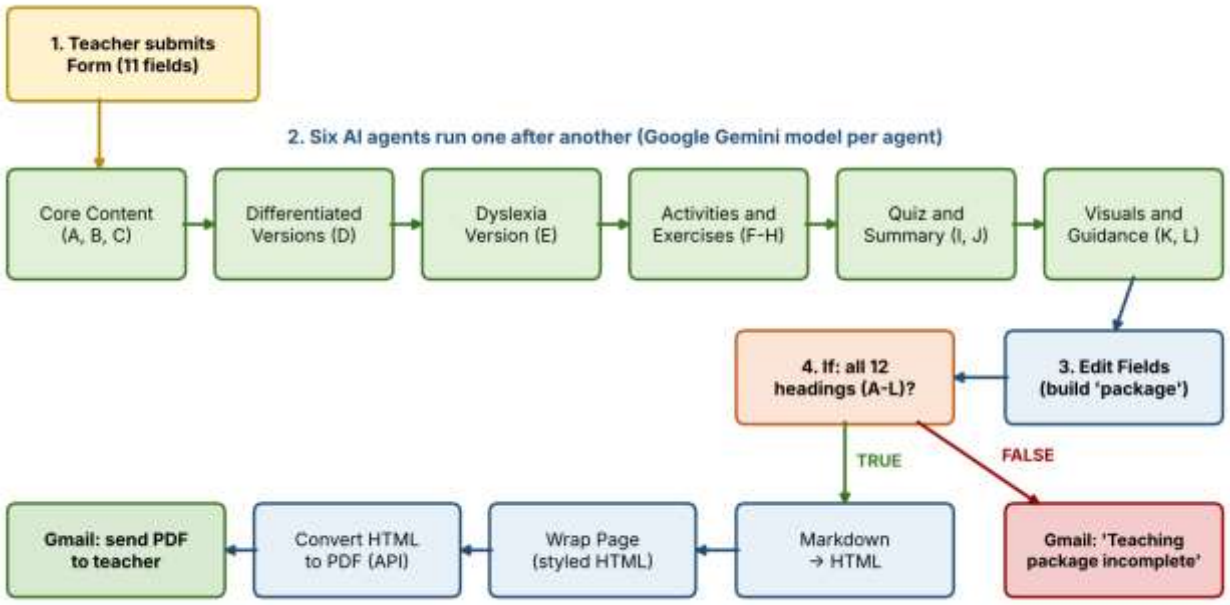


Figure 1: Workflow map from form submission to email delivery or failure alert

3.3 Process decomposition

The work is split across six agents so that each one has a focused task and a prompt of manageable size. The agents run one after another (not in parallel) to stay within model rate limits.

Agent	Sections	Technique	Why this technique
Core Content	A, B, C	Zero-shot	Clear task; no example needed
Differentiated Versions	D	One-shot (Magnets example)	One example fixes the structure and tone for three levels

Agent	Sections	Technique	Why this technique
Dyslexia Version	E	Few-shot (Magnets and Plants examples)	Examples show how the same concept is adapted
Activities and Exercises	F, G, H	Zero-shot	Clear checklist of required items
Quiz and Summary	I, J	Zero-shot	Clear checklist of required items
Visuals and Guidance	K, L	Zero-shot	Clear checklist of required items

Section	Title	Audience
A	Learning Objectives and Success Criteria	Teacher and students
B	Teacher Explanation	Teacher
C	Student Handout	Students
D	Differentiated Learning Versions (Beginner, Intermediate, Advanced / Challenge)	Students
E	Dyslexia-Friendly Learning Version	Students and teacher
F	Examples and Activities	Teacher and students
G	Practice Exercises	Students
H	Discussion Questions	Teacher and students
I	Quiz and Answer Key	Students; key for teacher
J	Summary	Students and teacher
K	Visual-Content Prompts	Teacher
L	Accessibility and Teacher Guidance	Teacher

3.4 Integrations and error-handling plan

Risk	Planned control
One or more sections missing from the AI output	If node checks for all 12 headings; on failure a notification email is sent and no PDF is produced
Temporary AI or rate-limit error	Retry On Fail is switched on for all six agents (n8n default of 3 tries). Wait between tries is 1000 ms (1 s); the specification suggests 10 s, see Section 14.
Wrong node name breaks an expression	Node names must not be changed (Section 12.3)
Gmail or PDF service failure	Check the Executions tab for the red node and its error message (Section 11)
Hallucinated facts or standards	System message forbids invented standards; teacher review required; email reminder

4. How It Works

The complete workflow as it appears in the n8n editor is shown below. Data moves left to right through the six agents, then into the validation step, which branches to the PDF path (upper) or the failure notification (lower).



Figure 2: Complete workflow in the n8n editor (20 nodes)

4.1 Node inventory

#	Node name	Type	Role
1	On form submission	Form Trigger	Collects the teacher's inputs
2	Core Content	AI Agent	Sections A, B, C
3	Differentiated Versions	AI Agent	Section D
4	Dyslexia Version	AI Agent	Section E
5	Activities and Exercises	AI Agent	Sections F, G, H
6	Quiz and Summary	AI Agent	Sections I, J
7	Visuals and Guidance	AI Agent	Sections K, L
8-13	Google Gemini Chat Model to Chat Model5	Chat Model	One model sub-node per agent
14	Edit Fields	Set	Joins the six outputs into one field, package
15	If	Condition	Checks that all 12 headings exist
16	Markdown	Markdown	Converts package text to HTML
17	Wrap Page	Set	Wraps the HTML in a styled page (field page)
18	Convert HTML to PDF	HTTP Request	Calls the PDF conversion API
19	Send a message	Gmail	Sends the PDF (success path)
20	Send a message1	Gmail	Sends the failure message (false branch)

4.2 Step-by-step walkthrough

Step 1: On form submission (Form Trigger)

Starts the workflow when a teacher submits the form. Title: Teaching Materials Generator. Description: Enter a topic once and get a full resource package by email. Authentication: None. Test URL is used while building; the Production URL is shared with teachers after the workflow is published.

Field	Required
Subject	Yes
Grade Level	Yes
Lesson Topic	Yes
Learning Objective	Yes
Lesson Duration	Yes
Curriculum or Standards	No
Language	Yes
Available Resources	No
Specific Learner Needs	No
Assessment Requirements	No
Email	Yes

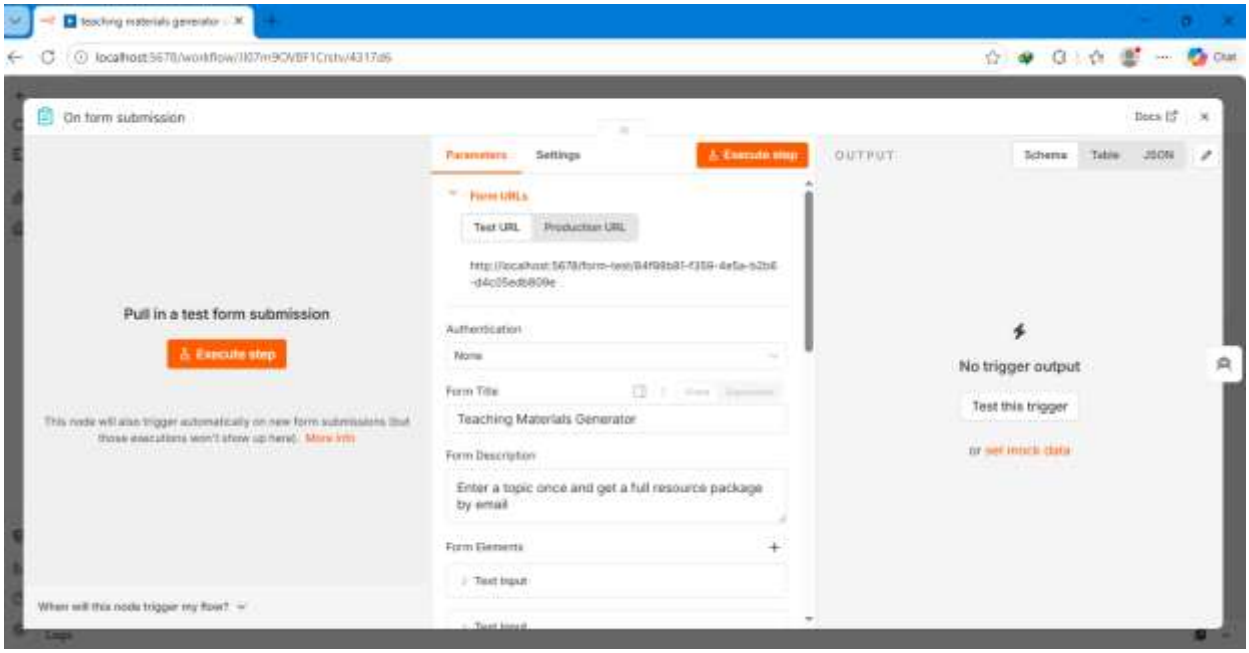


Figure 3: On form submission node: form URLs, title, description and elements

Steps 2 to 7: The six AI Agents

All six agents are configured the same way: Source for Prompt = Define below; the Prompt (User Message) is an expression that first lists the Teacher Inputs block and then the task text; the System Message is the shared system message. Each agent reads the form values by node name, for example {{ $('On form submission').item.json['Lesson Topic'] }}. The full prompt text is in Section 6.

Google Gemini Chat Model. Each agent has its own Gemini Chat Model sub-node (credential: Google Gemini (PaLM) API credential; model: models/gemini-3.1-flash-lite). All six models use Sampling Temperature 0.3 for consistent, accurate science content.

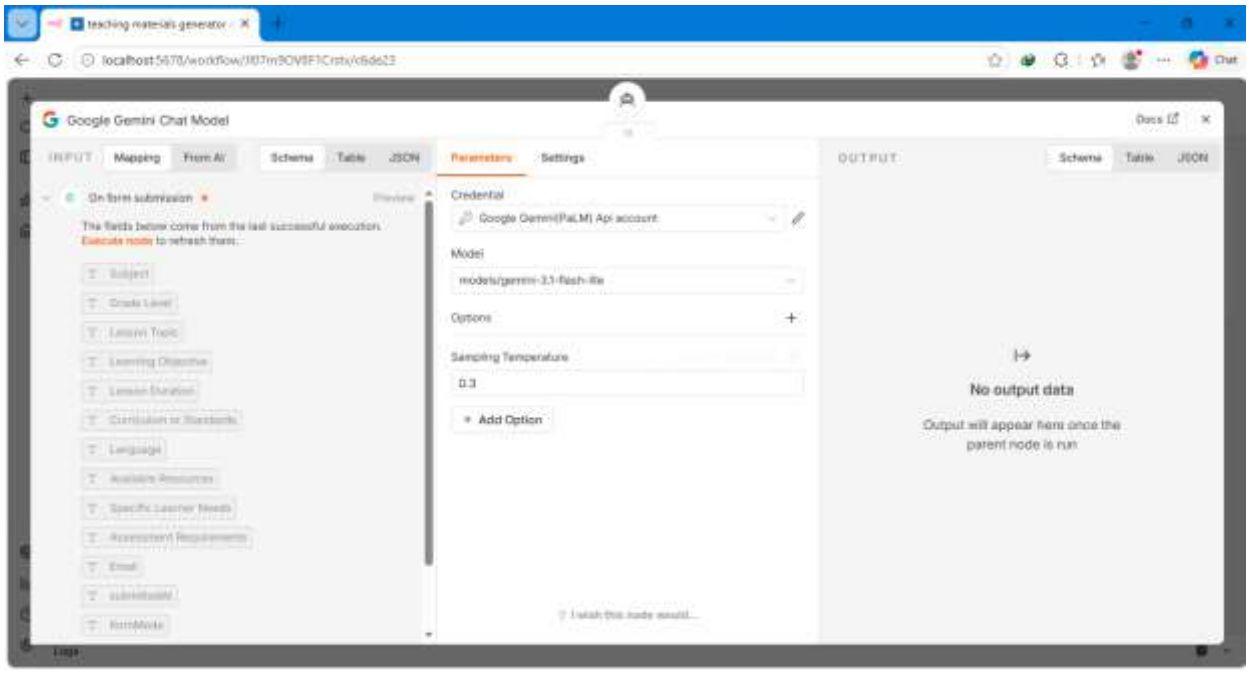


Figure 4: Google Gemini Chat Model: credential, model and sampling temperature

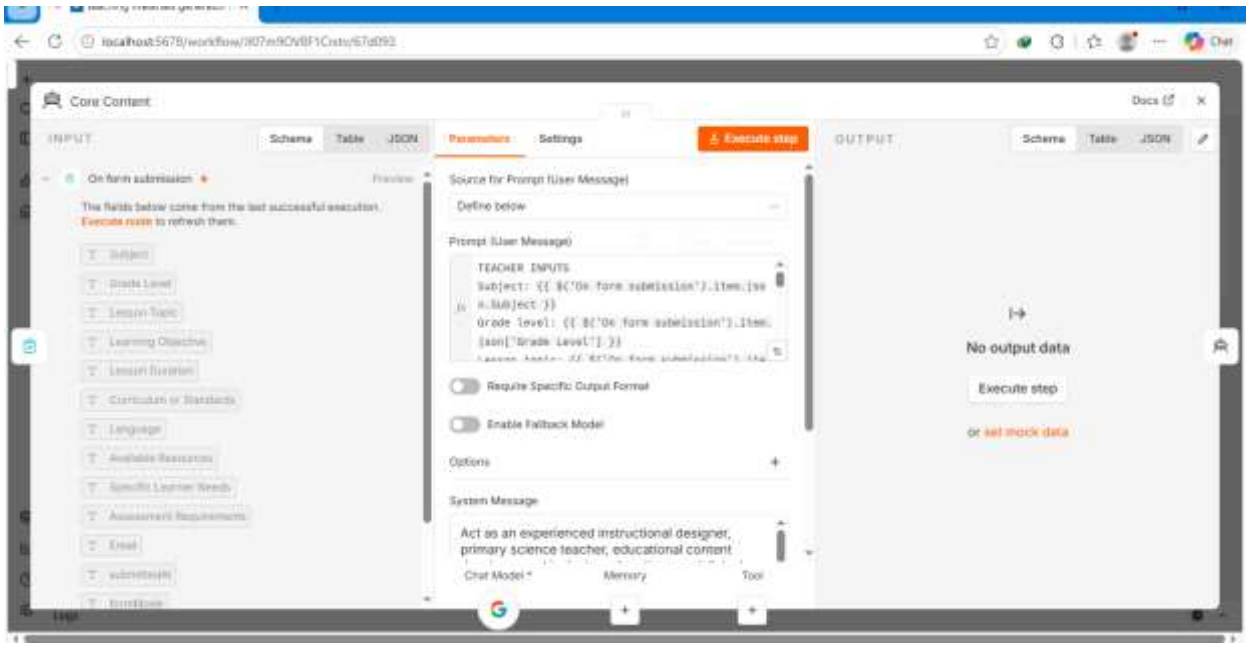


Figure 5: Core Content agent (sections A, B, C)

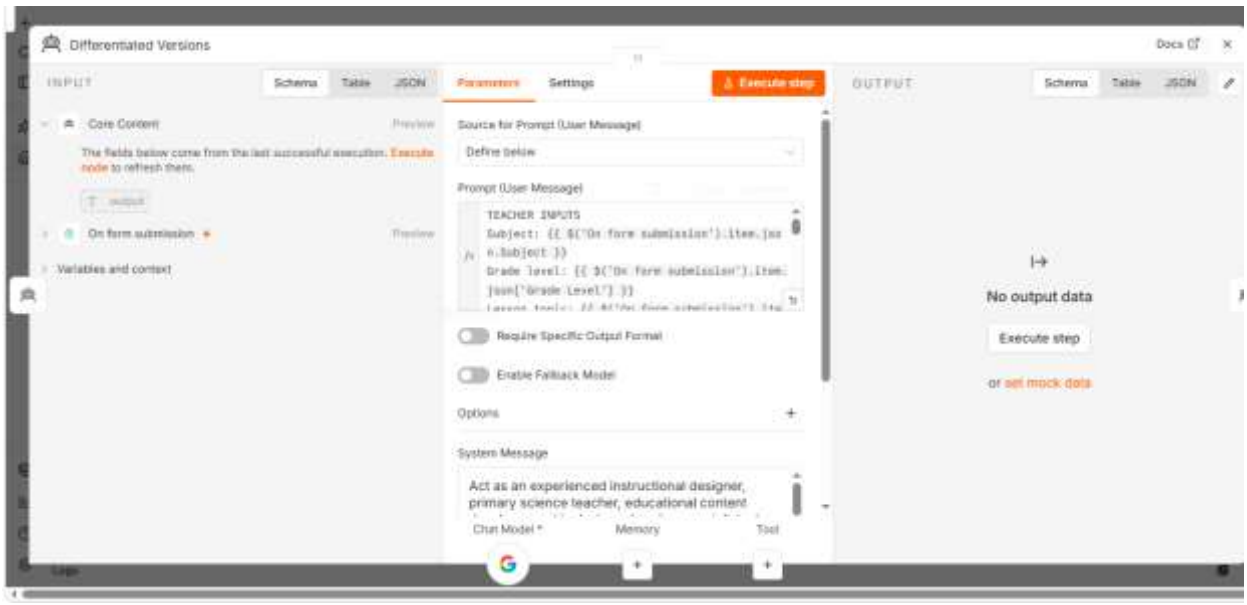


Figure 6: Differentiated Versions agent (section D)

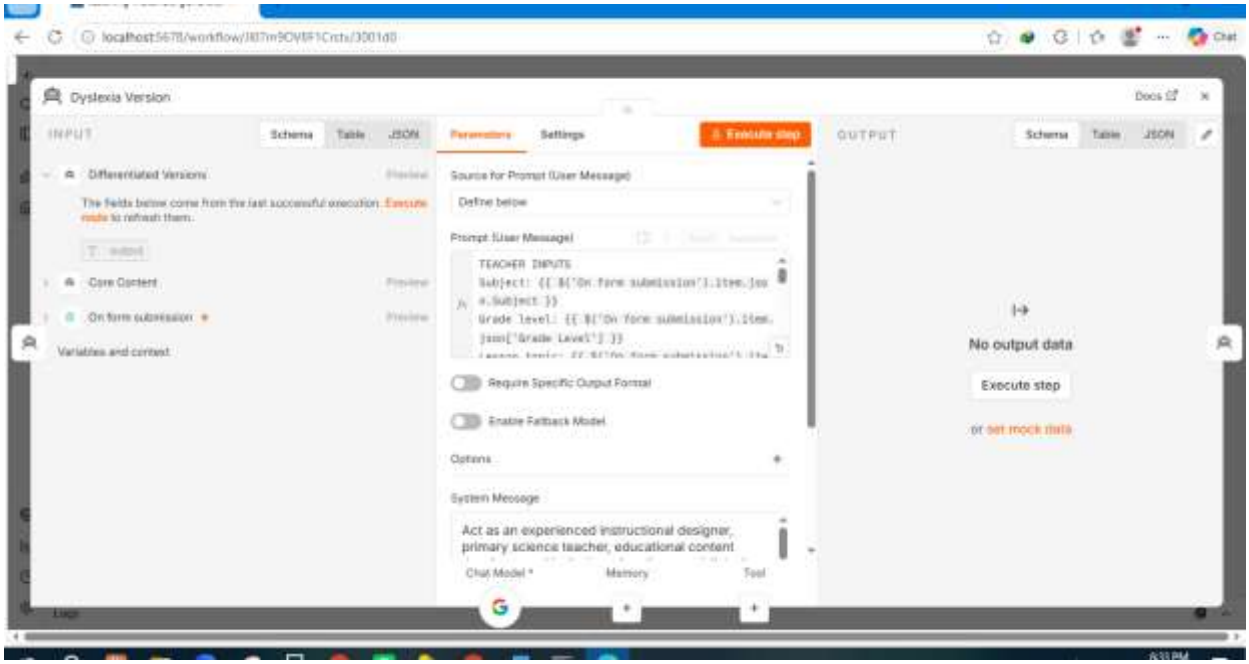


Figure 7: Dyslexia Version agent (section E)

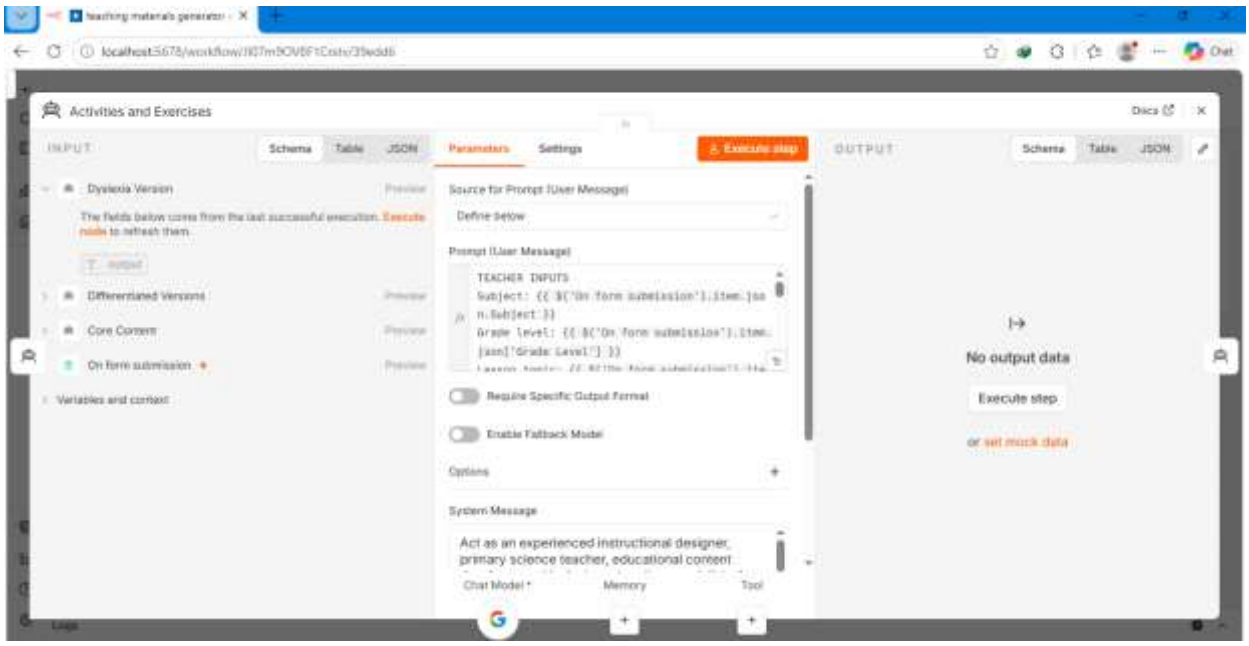


Figure 8: Activities and Exercises agent (sections F, G, H)

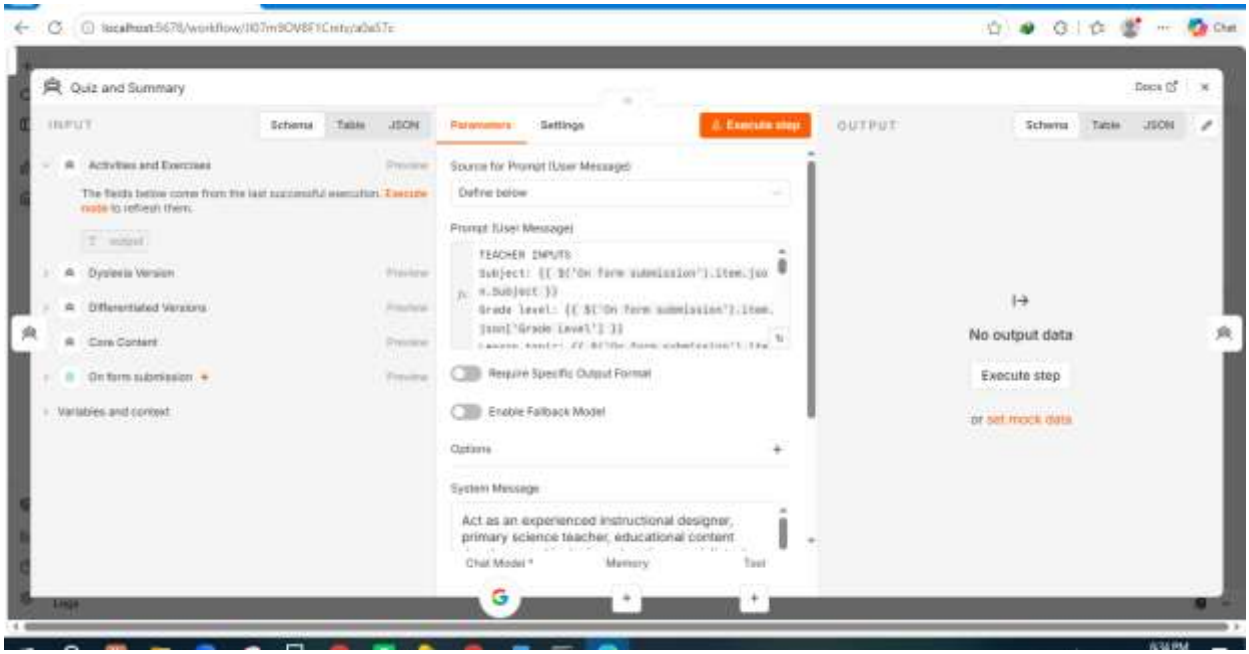


Figure 9: Quiz and Summary agent (sections I, J)



Figure 10: Visuals and Guidance agent (sections K, L); the message in the output panel confirms Retry On Fail is active

Retry On Fail is switched on in the Settings tab of every agent (Max. Tries 3, Wait Between Tries 1000 ms, On Error: Stop Workflow).

Important distinction: the If node sends an email only when the workflow reaches validation and one or more headings are missing. Because each AI Agent is configured with On Error: Stop Workflow, an agent execution error may stop the run before the If node and therefore may not trigger the failure email. Configure an Error Workflow or explicit error-handling path if owner notifications are required for these failures.

Step 8: Edit Fields (build the package)

Mode: Manual Mapping. Creates one string field named package that combines a title and info line (topic, subject, grade, duration, language) with the output of the six agents in order. This single field is what the validation and conversion steps read. Include Other Input Fields is OFF.

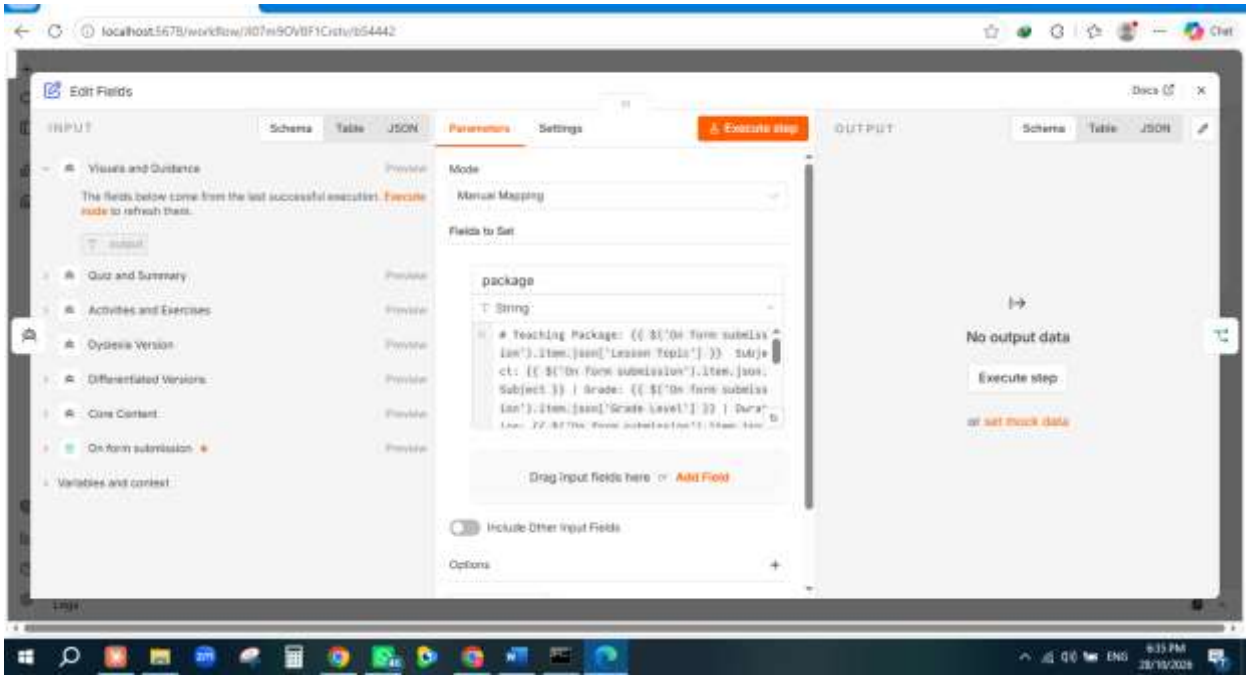


Figure 11: Edit Fields node: the package field

Step 9: If (validation check)

Checks that the package contains all twelve section headings. If every heading is found the item goes to the true output (Markdown). If any heading is missing it goes to the false output (failure email).

Condition (expression)

```
{{ ['## A.','## B.','## C.','## D.','## E.','## F.','## G.','## H.','## I.','## J.','## K.','## L.']
   .every(h => $json.package.includes(h)) }}   -> Boolean, is true
```

Limitation: the check confirms that headings exist. It does not judge whether the content is correct, which is why human review is still required.

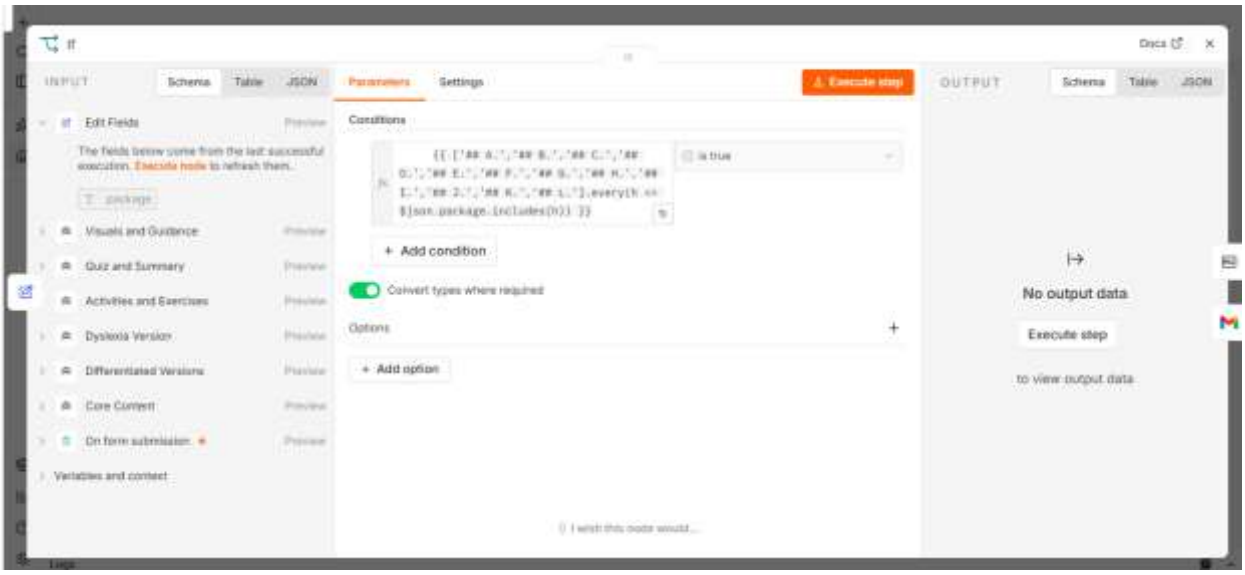


Figure 12: If node: condition that tests for all 12 section headings

Step 10: Markdown

Mode: Markdown to HTML. Input: {{ $json.package }}. Destination key: html. Tables support is ON, so tables in the AI output are converted correctly.

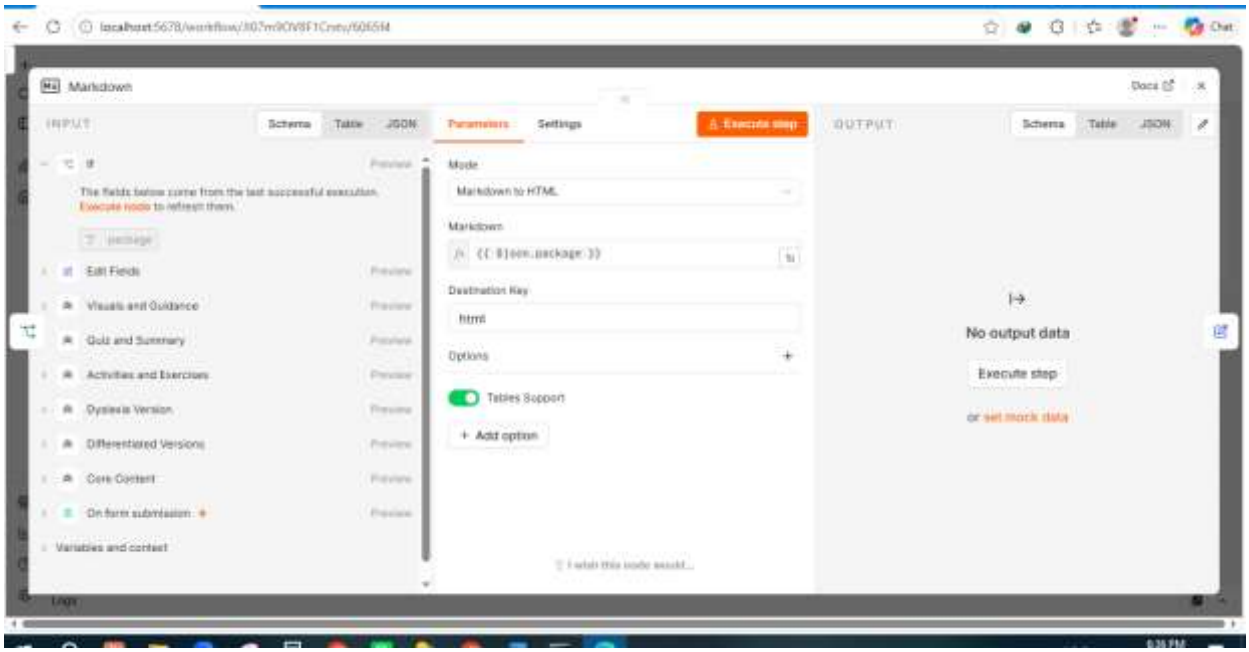


Figure 13: Markdown node: package converted to the html field

Step 11: Wrap Page

Wraps the HTML in a complete page with simple styling and writes it to a field named page. The sans-serif font (Arial), 14 px size and 1.6 line height follow readability guidance for dyslexia-friendly material.

Value of the page field

```
<html><head><meta charset="utf-8"><style>body{font-family:Arial,sans-serif;font-size:14px;line-
height:1.6;marg
in:30px} h1,h2,h3{margin-top:24px} table{border-collapse:collapse} td,th{border:1px solid
#999;padding:6px}</s
tyle></head><body>{{ $json.html }}</body></html>
```

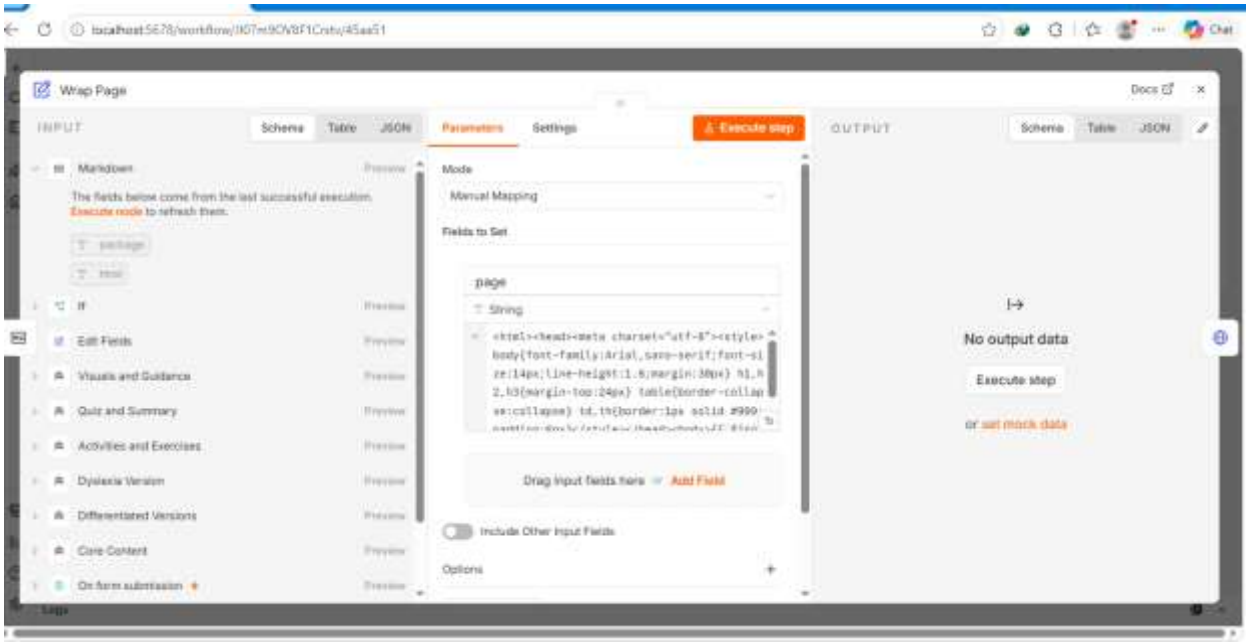


Figure 14: Wrap Page node: styled HTML page

Step 12: Convert HTML to PDF (HTTP Request)

POST request to the html2pdf.app API. The body is JSON containing the page HTML. The response format is set to File, so the PDF is returned as a binary attachment for the next node. The key is entered as the apiKey query parameter (collapsed in the screenshot).

Configuration

```
Method: POST
URL: https://api.html2pdf.app/v1/generate
Send Query Parameters: ON   apiKey = [HIDDEN]
Send Body: ON   Body Content Type: JSON
JSON Body: { "html": {{ JSON.stringify($json.page) }} }
Options > Response > Response Format: File
```

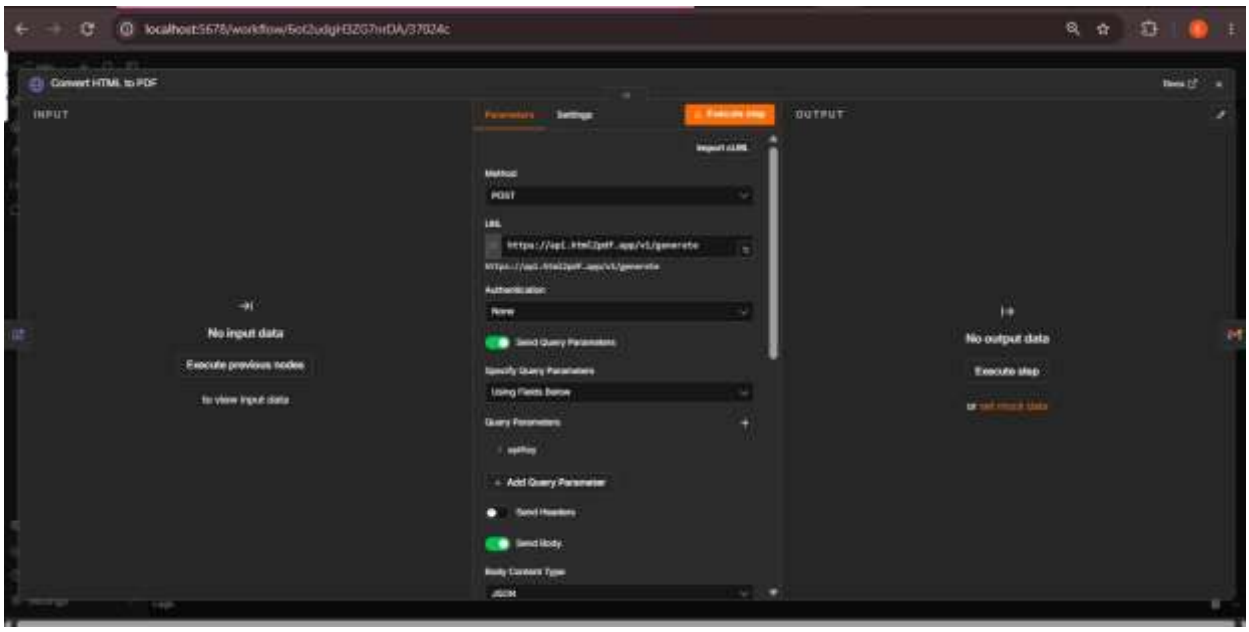


Figure 15: Convert HTML to PDF node: the API key is entered as the apiKey query parameter

Step 13: Send a message (Gmail, success path)

Sends the PDF to the teacher. To: the Email value from the form. Subject: Teaching package: [Lesson Topic] (please review before use). Sender name: TeachWise assistant. Email type: Text. The attachment is the binary field data from the previous node. The message tells the teacher that a human check is required before use with students. Credential: Gmail OAuth2.

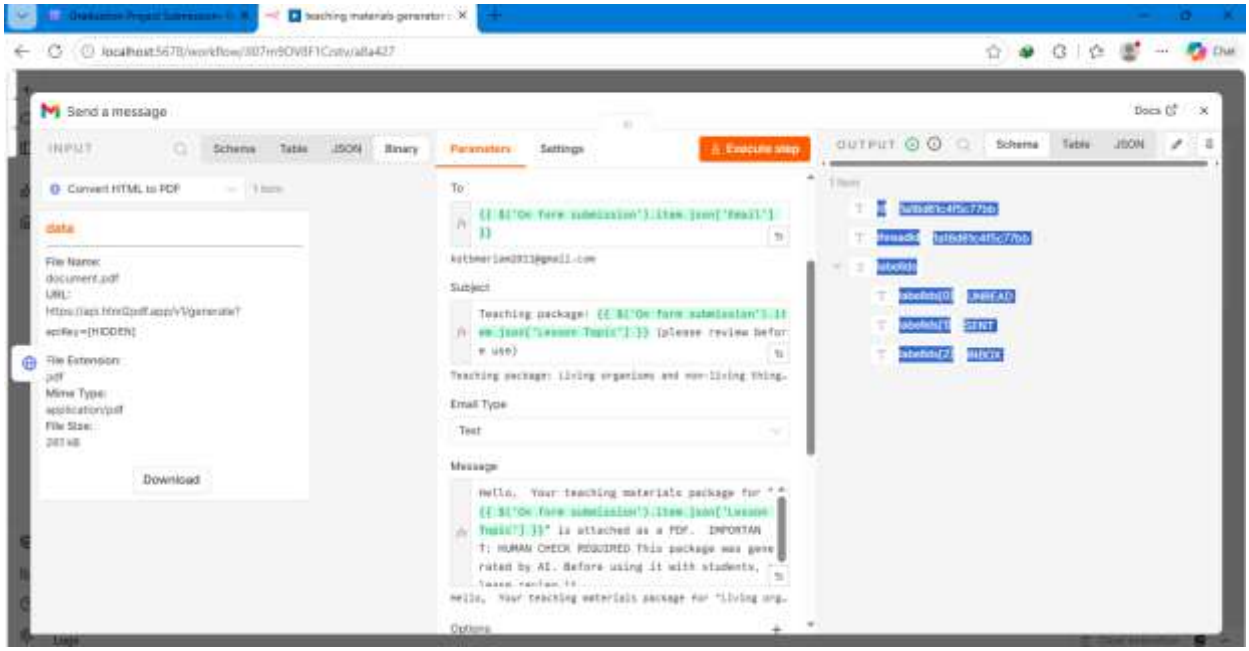


Figure 16: Send a message node: success email with the PDF attached

Step 14: Send a message1 (Gmail, failure path)

Runs only when the If node finds a missing section. Subject: Teaching package incomplete. Message: "A section is missing. Please run the form again." The recipient is the Email value entered in the form, so the person who submitted it is told to run it again. The same Email field also shows who submitted each request.

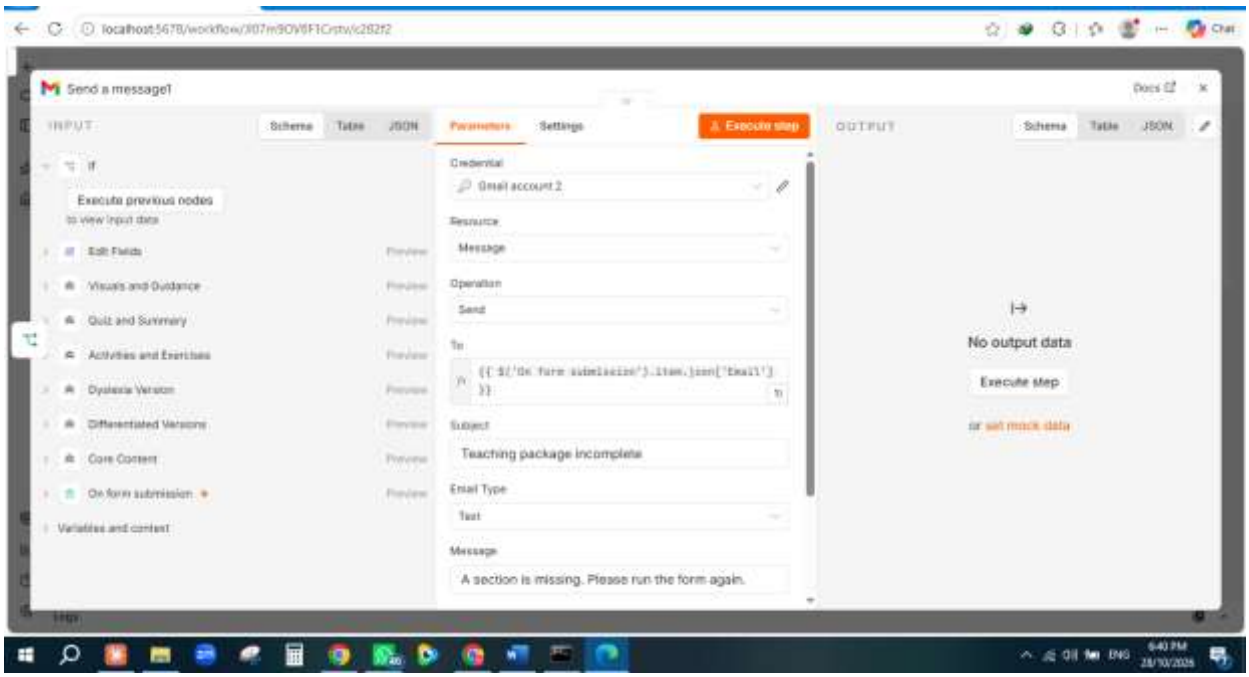


Figure 17: Send a message1 node: failure notification

4.3 Result

The teacher receives an email titled "Teaching package: [topic]" with the PDF attached. The screenshot shows a delivered package for "Living organisms and non-living things".

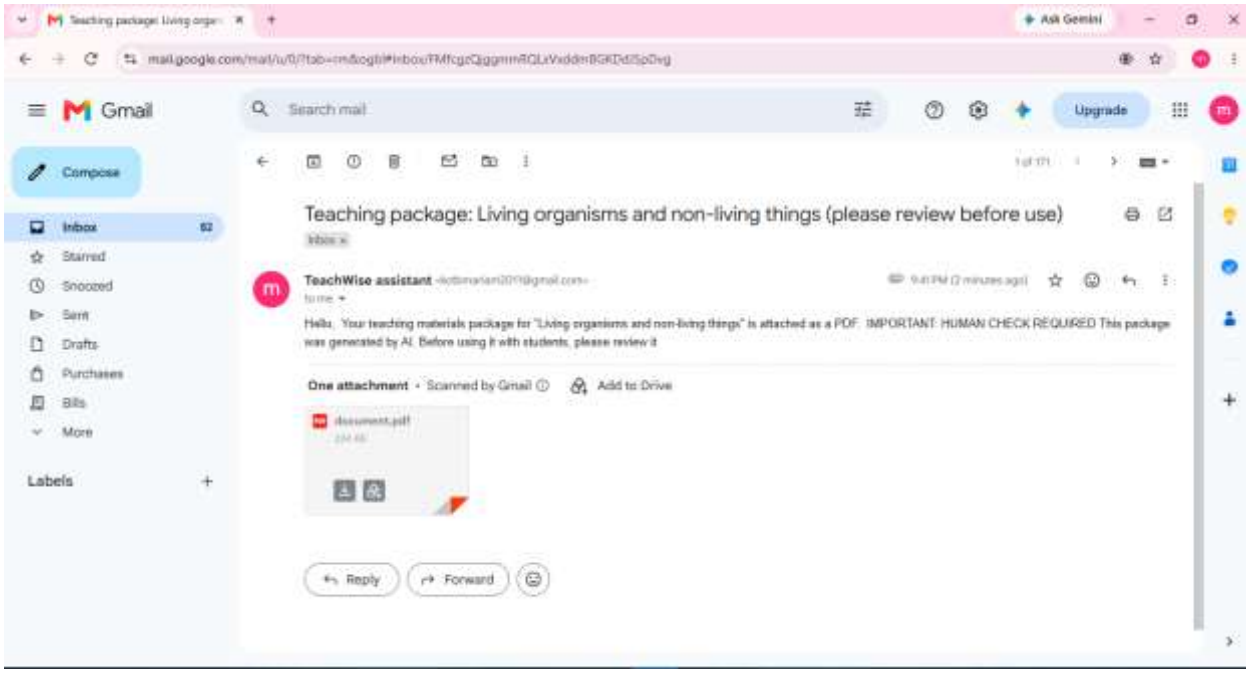


Figure 18: Delivered email with the PDF attachment

5. Integration Details

Integration	Used by	Setup	Notes
Google Gemini (PaLM) API	6 Gemini Chat Model nodes	API key from Google AI Studio, entered in an n8n Google Gemini (PaLM) API credential; default host https://generativelanguage.googleapis.com	Model: models/gemini-3.1-flash-lite. Free-tier rate limits apply.

Integration	Used by	Setup	Notes
Gmail OAuth2	Send a message, Send a message1	Gmail OAuth2 credential. Google Cloud project with Gmail API enabled, OAuth client (Web application), redirect URI http://localhost:5678/rest/oauth2-credential/callback, sender listed under Test users	While the Google Cloud app is in Testing status, authorisation may need to be renewed periodically (Section 11).
html2pdf.app	Convert HTML to PDF	API key supplied as the apiKey query parameter, stored as plain text in the node	Free quota applies. Recommended: move the key into n8n credentials (Section 14).
ngrok	Public access to the form	ngrok http 5678, then start n8n with WEBHOOK_URL set to the ngrok address	The PC, n8n and ngrok must stay running while teachers use the form.

5.1 Rate limits and quotas

- Gemini: free-tier limits can slow or block runs if many teachers submit at the same time. Check usage in Google AI Studio.
- html2pdf.app: the free plan has a monthly conversion quota. Check usage in its dashboard.
- Gmail: standard sending limits of the sender account apply.

6. AI System Prompts

This section contains the full text of every prompt used, copied from the exported workflow. Each prompt follows the production structure from the course: Role, Task, Constraints, Output format (exact headings), and edge-case rules.

Element	Where it is defined
Role	Shared system message: instructional designer, primary science teacher, content developer, inclusive education specialist
Rules and constraints	Shared system message: write in the teacher's language; state assumptions for blank inputs; never invent curriculum standards; separate teacher and student content; label audience and level
Output format	Task text: exact section headings (## A. to ## L.), which the If node later checks
Edge cases	Blank input: state a reasonable assumption in one line. No curriculum entered: do not claim alignment.
Self-check	Shared system message: check facts, grade suitability and answer correctness before finishing

6.1 Shared system message (all six agents)

```
Act as an experienced instructional designer, primary science teacher, educational content developer
and
inclusive education specialist who knows differentiated instruction, Universal Design for Learning
(UDL)
and strategies for supporting students with dyslexia. The learners are primary-school students.
Write in
the language given in the teacher inputs. If an input is blank, state a reasonable assumption in one
```

```
line. Never invent curriculum standards or claim alignment that was not verified. Keep teacher
guidance
separate from student-facing content, and label each resource with its audience and learner level.
Use
exactly the headings requested. Before finishing, check that facts are accurate, content suits the
grade,
and answers match the questions. Keep every section concise.
```

6.2 Teacher Inputs block (start of every agent prompt)

```
TEACHER INPUTS
Subject: {{ $('On form submission').item.json.Subject }}
Grade level: {{ $('On form submission').item.json['Grade Level'] }}
Lesson topic: {{ $('On form submission').item.json['Lesson Topic'] }}
Learning objective: {{ $('On form submission').item.json['Learning Objective'] }}
Lesson duration: {{ $('On form submission').item.json['Lesson Duration'] }}
Curriculum or standards: {{ $('On form submission').item.json['Curriculum or Standards'] }}
Language: {{ $('On form submission').item.json.Language }}
Available resources: {{ $('On form submission').item.json['Available Resources'] }}
Specific learner needs: {{ $('On form submission').item.json['Specific Learner Needs'] }}
Assessment requirements: {{ $('On form submission').item.json['Assessment Requirements'] }}
```

6.3 Agent 1: Core Content (sections A, B, C), zero-shot

Task text that follows the Teacher Inputs block:

```
Write these sections using these exact headings:

## A. Learning Objectives and Success Criteria
- Clear, measurable objectives; what students should know or be able to do
- Simple "I can..." success criteria students understand
- The same core objective applies to every differentiated version

## B. Teacher Explanation (teacher-facing)
- Accurate, age-appropriate explanation in manageable steps
- Essential vocabulary with simple definitions
- Relatable examples and analogies
- Common misconceptions and how to address them

## C. Student Handout (student-facing)
- Short introduction, key concepts, examples, brief review
- Headings, bullet points, numbered steps, clear instructions
- No unnecessary information or excessive text
```

6.4 Agent 2: Differentiated Versions (section D), one-shot

Task text that follows the Teacher Inputs block:

```
Write section D using the same structure, tone and quality as this teacher-approved example.

EXAMPLE (topic: Magnets)
### Beginner (student-facing)
Magnets pull on some metals. Try it: touch a magnet to a paper clip. Does it stick? Circle: YES /
NO. Hint: look at the picture of the paper clip.
### Intermediate (student-facing)
Magnets attract objects made of iron, nickel or cobalt. Predict which 4 classroom objects a magnet
will pick up. Test them. Explain one result in a full sentence.
### Advanced / Challenge (student-facing)
Explain why a magnet picks up a steel nail but not an aluminium can. Design a fair test for 5
materials and say what you will keep the same.
NOW WRITE THIS FOR THE LESSON ABOVE, using these exact headings:

## D. Differentiated Learning Versions
```

```
### 1. Beginner (short explanations, familiar words, worked example, visual supports, hints, small
steps, frequent checks for understanding)
### 2. Intermediate (grade-level explanation, independent practice, mix of
recall/comprehension/application, explain reasoning)
### 3. Advanced / Challenge (problem-solving, analysis, investigation, extension; compare, justify,
predict or design)
```

6.5 Agent 3: Dyslexia Version (section E), few-shot

Task text that follows the Teacher Inputs block:

```
Write section E. Use the two examples below to see how the SAME concept is adapted for each level,
including the dyslexia-friendly version.

EXAMPLE 1 (topic: Magnets)
Beginner: Magnets pull on some metals. Touch a magnet to a paper clip. Does it stick?
Intermediate: Magnets attract iron, nickel and cobalt. Predict which objects a magnet picks up, then
test.
Advanced: Explain why a magnet picks up a steel nail but not an aluminium can.
Dyslexia-friendly: Key words: **magnet**, **attract**. Word bank: magnet | metal | stick. Steps: 1.
Hold the magnet near a paper clip. 2. Does it stick? 3. Say your answer out loud or draw it.
Sentence starter: "The magnet sticks to ___."

EXAMPLE 2 (topic: Plants need light)
Beginner: Plants need light to grow. A plant in a dark box gets weak.
Intermediate: Plants use light to make food. Predict what happens to a plant kept in the dark for
one week.
Advanced: Design an experiment to test how different amounts of light change plant growth.
Dyslexia-friendly: Key words: **light**, **grow**. Word bank: plant | light | dark. Steps: 1. Look
at plant A (light). 2. Look at plant B (dark). 3. Draw both plants. Sentence starter: "The plant in
the dark looks ___."

NOW WRITE THIS FOR THE LESSON ABOVE, using these exact headings:

## E. Dyslexia-Friendly Learning Version
### Vocabulary Preview and Word Bank (new words in small groups)
### Short Explanation (concise sentences, short sections with descriptive headings, key terms in
bold, not colour alone)
### Guided Example (before independent work)
### Student Tasks (numbered sequential steps, short task sections, sentence starters)
### Ways to Show Understanding (oral answers, diagrams, sorting, practical demonstration)
### Adaptations Used (extra processing time, text-to-speech or audio, no penalty for spelling or
decoding errors, clear layout, sans-serif font in the final document, and the purpose of each)

Treat the strategies as adaptable options, not as the needs of every student with dyslexia. Keep the
same scientific content and learning expectations as the other versions.
```

6.6 Agent 4: Activities and Exercises (sections F, G, H), zero-shot

Task text that follows the Teacher Inputs block:

```
Write these sections using these exact headings:

## F. Examples and Activities
- At least 3 age-appropriate examples
- Guided practice and independent practice
- One interactive or hands-on activity (use the available resources)
- One visual or multisensory activity
- Short adaptation notes for Beginner, Intermediate, Advanced and Dyslexia-friendly

## G. Practice Exercises (student-facing, clear instructions)
Include: multiple-choice, matching or sorting, fill-in-the-blank with a word bank, short-answer, and
one application or problem-solving task.

## H. Discussion Questions
```

```
Include: simple comprehension, "explain your thinking", real-life application, and extension
questions for advanced learners. Add optional sentence starters and oral-response alternatives.
```

6.7 Agent 5: Quiz and Summary (sections I, J), zero-shot

Task text that follows the Teacher Inputs block:

```
Write these sections using these exact headings:

## I. Quiz and Answer Key
- Quiz: 6 questions in mixed types, matching the learning objective and the assessment requirements.
Do not make spelling, reading speed or handwriting the focus.
### Answer Key (teacher-facing)
- Correct answers, each with a brief explanation
- A simple scoring guide
- Accessible presentation and response options (read aloud, oral answers, drawing, extra time)

## J. Summary
- A concise lesson summary
- 3 to 5 key takeaways
- A short student-friendly recap
- An optional vocabulary review
```

6.8 Agent 6: Visuals and Guidance (sections K, L), zero-shot

Task text that follows the Teacher Inputs block:

```
Write these sections using these exact headings:

## K. Visual-Content Prompts
Write specific, ready-to-use prompts for: (1) a labeled diagram, (2) a step-by-step concept visual,
(3) sorting cards or a visual organizer, (4) a simple infographic. Each must be scientifically
accurate, age-appropriate and uncluttered. Add a note that essential information must also be in
text.

## L. Accessibility and Teacher Guidance (teacher-facing)
- The adaptations used in each version
- The purpose of each dyslexia-support strategy
- Ways to monitor student understanding
- Options for extra scaffolding or extension
- Limitations and anything the teacher should verify before use (including any assumptions made and
standards alignment)
```

Prompt versions and changes must be recorded in the change log (Section 12.3). A fuller discussion of each prompt is in the separate Prompt Library document.7. Testing and Results

7.1 Testing approach

Level	How it applies to this workflow
Unit testing	Run each AI Agent alone with Execute step, using pinned form data, and check its headings and content.
Integration testing	Submit the form and follow the whole run in the Executions tab until the email arrives.
Edge-case testing	Leave optional fields blank; use a different grade; force a missing section to test the failure branch.

7.2 Recorded test run

Item	Detail
Date	30 October 2026
Inputs	Lesson topic: Living organisms and non-living things
Result	All nodes completed; email "Teaching package: Living organisms and non-living things (please review before use)" received with an attached PDF (234 KB)
Sample package	A complete example package (9 pages, Science, Grade 4, 45 minutes, English) is provided as a separate PDF
Status	**Passed**

7.3 Test plan

Test	Input	Expected result	Status
Normal run	Science, Grade 4, topic, objective, 45 min, English, valid email	All nodes green; PDF arrives by email	**Passed**
Different level	Same topic at Grade 1 and Grade 5	Language, support and difficulty differ clearly	**To run**
Optional fields blank	Standards, resources, learner needs, assessment left empty	Package created; assumptions stated in the content	**To run**
Missing section	Remove one agent's output (testing only)	If goes to false branch; failure email arrives	**To run**
Invalid email	Wrong address in the form	Form blocks it or Gmail node shows an error	**To run**
Public link	Open the Production URL in a private window	Form opens (after ngrok warning page) and starts a run	**To run**

8. Pre-Launch Checklist

Check	Status	Evidence / action
All integrations connected and tested	**Done**	Gemini, Gmail and PDF service all worked in the test run
Error handlers in place for critical paths	**Partly**	Validation branch and Retry On Fail on all six agents. No global Error Workflow is configured.
Secrets stored securely (not hard-coded)	**Fix**	The html2pdf key is stored as plain text in the node. Move it to a credential and regenerate the key.
Teams know what to expect	**To do**	Tell teachers the run takes several minutes and that content must be reviewed
Someone assigned to monitor in the first week	**To do**	Name the person in Section 13
Documentation written	**Done**	This document, the Technical Workflow Specification and the Prompt Library
Workflow marked Active	**Done**	Exported file shows active: true
Backup plan if automation fails	**Defined**	Teacher prepares materials manually, or the form is resubmitted once the issue is fixed
Cost monitoring set up	**To do**	Check Gemini usage in Google AI Studio and html2pdf quota monthly
Rollback plan documented	**Defined**	Re-import the previous exported JSON (Section 12.3)

9. Monitoring and KPIs

9.1 What to monitor

Area	What to watch	Where
Execution log	Every run, its duration and any red node	n8n, Executions tab
Error rate	Failed executions and failure-branch runs (target: under 1% of runs fail unexpectedly)	Executions tab; failure emails
Response time	A run normally takes several minutes because six model calls run in sequence. Investigate runs that are much longer.	Execution duration
Quota and cost	Gemini request quota; html2pdf conversion quota	Google AI Studio; html2pdf dashboard
Quality	Teacher feedback on accuracy and usefulness; classification of recurring problems	Feedback from teachers

9.2 Key performance indicators

KPI	How to measure	Baseline	Current
Time saved	Manual preparation time per lesson minus time to review the generated package	To be measured	To be measured
Accuracy	Share of generated packages that teachers approve without factual corrections	Not applicable	To be measured
Reliability	Successful runs divided by total runs	Not applicable	1 of 1 recorded run
Cost	Gemini usage plus PDF service plan per package	Not applicable	Free tiers in use
Throughput	Packages generated per week	Not applicable	To be measured

9.3 Recommended rollout (shadow mode)

1. Phase 1: a small group of teachers uses the packages but reviews every section before use.
2. Phase 2: wider use; teachers report corrections; owner updates prompts as needed.
3. Phase 3: routine use with spot checks; monitor failure rate and quota.

10. Runbook

10.1 Start the system

1. Open a Command Prompt and run `ngrok http 5678`. Keep the window open and note the Forwarding address (https://...ngrok-free.dev).
2. Open a second Command Prompt and run: `set WEBHOOK_URL=https://YOUR-ADDRESS.ngrok-free.dev/`, then `set N8N_PROXY_HOPS=1`, then `n8n start`.
3. Open n8n (http://localhost:5678) and make sure the workflow is published (Active).
4. Open the On form submission node, select the Production URL tab and copy the link.
5. Send the link only to trusted teachers. On first visit ngrok may show a warning page; teachers click Visit Site.

10.2 Normal operation

- Keep the PC, the ngrok window and the n8n window running while the form is in use.
- If ngrok gives a different address, repeat steps 2 to 5 above, or reserve a fixed free domain in the ngrok dashboard.
- After any change to the workflow: save, then publish again so the live form uses the new version.

10.3 Stop the system

- Close the ngrok window to stop public access immediately.
- Unpublish (deactivate) the workflow in n8n to stop new runs.
- Close the n8n window to shut down.

10.4 Daily and weekly checks

- Daily (first week): open Executions, confirm recent runs are green, and check for failure emails.
- Weekly: review quota in Google AI Studio and html2pdf; confirm Gmail authorisation is still valid.
- Monthly: review failure patterns, collect teacher feedback, review prompts and the model choice.

10.5 Escalation

Situation	First action	Escalate to
Run fails once	Check red node in Executions; run again	Workflow owner
Repeated failures or credential errors	Follow Section 11	Workflow owner, then n8n administrator
Suspected misuse of the public link	Close ngrok; unpublish workflow	Workflow owner immediately
Exposed API key	Delete the key at the provider and create a new one; update n8n credential	Workflow owner immediately

11. Troubleshooting Guide

Symptom	Likely cause	What to do
Gemini node fails with rate-limit or temporary error	Free-tier limit or temporary service issue	Wait and run again; Retry On Fail is already on, so if it keeps failing increase Wait Between Tries; check quota in Google AI Studio
Gmail node: authentication failed or credential error	OAuth authorisation expired or revoked (can happen while the Google Cloud app is in Testing status)	Reconnect the Gmail credential in n8n; confirm the sender is listed under Test users
Convert HTML to PDF returns an error or empty file	Invalid or exhausted API key, bad HTML, or no internet	Check the key and quota in the html2pdf dashboard; check the Wrap Page output
Teacher receives "Teaching package incomplete"	One or more agents did not output its headings (## A. to ## L.)	Run the form again; open the execution and check which agent's output is missing; review that agent's prompt
Expression error mentioning a node	A node was renamed	Restore the original node names listed in Section 4.1
Form link does not open	PC, n8n or ngrok is off, or the ngrok address changed	Restart per Section 10.1 and send the new Production URL
Content language or level is wrong	Form inputs unclear or prompt needs tuning	Check the Language and Grade Level inputs; refine the relevant agent prompt
Run takes very long	Six sequential model calls; rate limiting	Wait; check Executions for the slow node
PDF attachment is named document.pdf	File name not set	See Future Improvements (Section 14)

12. Security, Compliance and Change Management

12.1 Data handling and privacy

- Lesson inputs are sent to the Google Gemini API. On the free tier, Google states that content may be used to improve its products. Do not enter student names or personal data in the form.
- The teacher's email address is used only to send the package and is held in n8n execution data.
- Generated packages are not written to a separate application database by this workflow; however, package content and binary files may remain in n8n execution data according to the instance retention and pruning settings. They also remain in the recipient’s email. Confirm execution-data retention settings before describing the workflow as not storing generated content.

12.2 Access and secrets

- Anyone with the form link can start a run, which uses model quota and sends mail from the owner's Gmail account. Share the link only with trusted people.
- The same public address also reaches the n8n sign-in page. Use a strong n8n password.
- Never show API keys in screenshots or share them. If a key is exposed, delete it and create a new one. Credentials are not included in exported workflow files.
- The html2pdf key currently appears as plain text in the Convert HTML to PDF node and in the exported JSON, so the exported file should be treated as confidential until this is fixed.

12.3 Version control and change log

- Export the workflow after each change (three-dot menu, Download) and name files with a version, for example v1.0, v1.1.
- Keep a master list of prompt versions (Section 6 and the Prompt Library).
- Do not rename nodes: expressions read other nodes by name.
- Rollback: import the previous JSON file as a new workflow, recreate credentials if needed, and publish it.

Version	Date	Changed by	Reason / change
1.0	31 October 2026	Dr Doaa Hafez	Initial documentation of the exported workflow (final)

12.4 Human review

The AI can make factual mistakes. The teacher must check the facts, quiz answers and answer key before using the materials. No curriculum standards are invented: when none are entered, the model states its assumptions.

13. Support and Handover

13.1 Contacts

Role	Name	Contact
Workflow owner / creator	Mariam Kotb	kotbmariam2011@gmail.com
First-week monitor	Mariam Kotb	kotbmariam2011@gmail.com
n8n and ngrok administrator	Mariam Kotb	kotbmariam2011@gmail.com
Google (Gemini and Gmail) account owner	Esraa Ahmed	Esraaahmedhelmy78@gmail.com
html2pdf account owner	Mariam Kotb	kotbmariam2011@gmail.com

13.2 Knowledge-transfer plan

Week	Activity
Week 1	Training on the architecture, nodes and prompts (Sections 4 and 6)
Week 2	Hands-on troubleshooting using Section 11; run the edge-case tests in Section 7.3
Week 3	Shadow mode: operations team monitors while the creator supports
Week 4	Full ownership by the operations team; creator available for escalations

13.3 Handover package

- This document
- Workflow export: teaching materials generator final.json
- Technical Workflow Specification (n8n)
- Prompt Library
- Sample output: Teaching material package (PDF)
- Credentials: to be created by the new owner (not included in the export)

14. Known Gaps, Limitations and Future Improvements

14.1 Differences between the exported workflow and the written specification

These were found by comparing the exported JSON file with the Technical Workflow Specification. They should be resolved so that the live workflow matches the documents.

#	Area	Exported workflow	Specification says	Recommended action
1	Retry wait time	Retry On Fail ON, Max. Tries 3, Wait Between Tries 1000 ms (1 s)	ON, 3 tries, 10 s wait	Optional: set Wait Between Tries to 10000 ms on all six agents
2	PDF API key	Plain text in the apiKey query parameter	Entered in authentication settings	Move to an n8n credential; regenerate the key

Design decision (not a gap): the failure email goes to the address entered in the form, not to the owner. The Technical Workflow Specification says the owner, so the specification should be updated to match.

14.2 Limitations

- A run takes several minutes because six model calls run one after another.
- The validation checks that headings exist; it does not verify content quality or accuracy.
- Free-tier rate limits may slow or block runs when many teachers submit at once.
- The public link works only while the PC, n8n and ngrok are running.
- The PDF step depends on an external online service and a valid key.
- The prompts are written for primary science; other subjects may need prompt changes.

14.3 Future improvements

- Set a clear PDF file name, for example Teaching-package-[topic].pdf, instead of document.pdf.
- Add a global Error Workflow so any failed run alerts the owner.
- Move the workflow to an always-on host instead of a local PC with ngrok.
- Log each run (topic, grade, time, status) to a Google Sheet to measure KPIs automatically.
- Add a confidence or review step that flags weak sections, or retry only the missing section instead of failing the whole run.
- Collect teacher ratings through a short feedback form linked in the email.
- Support more subjects and an editable Word output as well as PDF.

Appendix A: Data Dictionary

Field	Source	Transformation	Destination
Subject, Grade Level, Lesson Topic, Learning Objective, Lesson Duration, Language	Form (required)	None; read by node name	Agent prompts; Edit Fields title line; email subject (topic)
Curriculum or Standards, Available Resources, Specific Learner Needs, Assessment Requirements	Form (optional)	None; blank means the model states an assumption	Agent prompts
Email	Form (required)	None	Gmail To field (success and failure emails); identifies the submitter
output (per agent)	AI Agent nodes	Markdown text with exact headings	Edit Fields
package	Edit Fields	Title, info line and six outputs joined in order	If (check); Markdown
html	Markdown node	Markdown converted to HTML (tables ON)	Wrap Page
page	Wrap Page	HTML wrapped with styling	Convert HTML to PDF
PDF binary (data)	Convert HTML to PDF	Returned as a file	Gmail attachment

Document Review Notes

Review scope: This review checked the supplied documentation for internal consistency, clarity, handover risks, and obvious editorial issues. The n8n workflow export, Technical Workflow Specification, Prompt Library, and sample PDF were not included in this upload, so the actual node settings, prompts, credentials, and generated output could not be independently verified.